higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING

(First Paper)

NQF LEVEL 3

13 NOVEMBER 2019

This marking guideline consists of 4 pages.

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NOTE TO MARKERS: This marking guideline serves as a guideline only. Alternative correct answers must be considered.

QUESTION 1: REINFORCED CONCRETE DETAILING

1.1
- It provides a better estimation of reinforcement steel requirements.
- It provides better stock management for reinforcement.
- It is very useful during the auditing of reinforcement.
- It can be used for reinforcement, cutting, bending and making a skeleton.
- It provides quality control at site as reinforcement is provided.
- It saves steel reinforcement material.
- Cutting and bending can be done at a factory and transported to site.
- It provides bench marks for quantity and quality requirements for reinforcement.

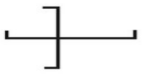
(Any THREE relevant answers) (3 × 2) (6)

1.2

16	Number of bars
R	Type
12	Rod size
05	Mark
150	Centre
T	Location/Comment

(6 × ½) (3)

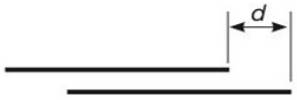
1.3 1.3.1



1.3.2



1.3.3



(3 × 2) (6)

1.4
- Plastic
- Steel
- Plywood
- Aluminium
- Fibreglass
- Solid timber

(Any 4 × 1) (4)

1.5

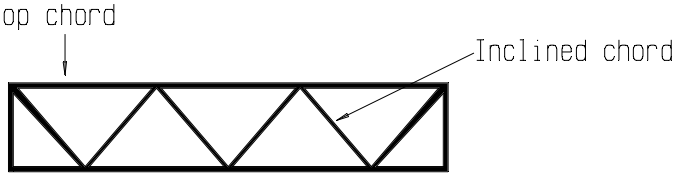
A	B	C	D	E	F	G	H				
Member								**Bending dimensions**			
	Bars per member	**No. of members**	**Total bars**	**Type/ diameter**	**Bar mark**	**Shape code**	**Total length**	**A**	**B**	**C**	**D**
Beam	**4** ✓	**4** ✓	**16** ✓	**Y12** ✓	**25** ✓	**32** ✓	**800** ✓	**500** ✓			
Beam	**6** ✓	**4** ✓	**24** ✓	**R8** ✓	**26** ✓	**32** ✓	**1 300** ✓	**900** ✓			

(16)
[35]

QUESTION 2: STRUCTURAL STEEL WORK DETAILING

2.1 A lattice girder is a frame with an open-web construction✓ which usually has parallel chords (and parallel booms)✓ when used for roofs and with internal web bracing members.✓ (3)

2.2



Correctness	2
Labelling	2
Line work	1

(5)

2.3
- IBR
- Super seal
- Kliplok (Any 2 × 1) (2)

2.4
- Black bolts — grade 4.6
- Black bolts — grade 8.8
- High-strength friction bolts (3)

2.5	2.5.1	H		
	2.5.2	D		
	2.5.3	C		
	2.5.4	B		
	2.5.5	G		
	2.5.6	F		
	2.5.7	E		
	2.5.8	J		
			(8 × 1)	(8)
2.6	2.6.1	Hold-down bolt		(1)
	2.6.2	A – H-section, I-section, column or stanchion B – Gusset plate D – Base plate		(3)
				[25]

QUESTION 3: WELDING

3.1	•Longitudinal •Transverse •Angular •Twisting •Bending			(5)
3.2	3.2.1	Fillet welded all-around in the factory		
	3.2.2	Fillet welded all-around on the construction site		
			(2 × 2)	(4)
3.3	•Carbon steel •High and low alloy steel •Stainless steel •Cast iron •Ductile iron		(Any 1 × 1)	(1)
3.4	• Butt joint • Lap joint • T-joint • Edge joint • Angle joint			(5)
				[15]
			TOTAL:	**75**

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